

E44 PCB Design Rules

LPKF ProtoMat E44 · all dimensions in mm

Tool set

Universal Cutter

0.2–0.5 mm

Isolation, width set to 0.3 mm.

End mill

1.0 mm

Copper rubout. Pour GND where you can.

Contour router

1.0 mm

Board outline, slots and cutouts.

Spiral drills

0.7–2.0 mm

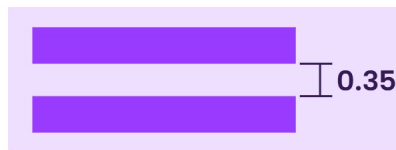
0.7, 0.8, 1.0, 1.5 and 2.0 mm.



Track width

0.5 mm

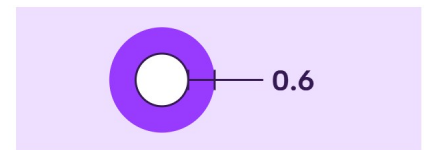
Minimum. Set as the net class default.



Clearance

0.35 mm

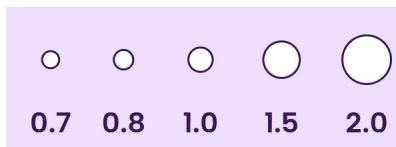
Copper to copper, pads in a footprint included.



Annular ring

0.6 mm

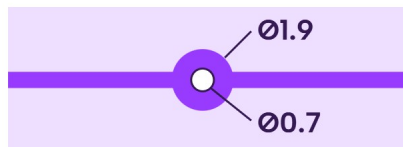
Minimum. 0.5 mm allowed on 1.0 mm holes.



Drill sizes

0.7–2.0 mm

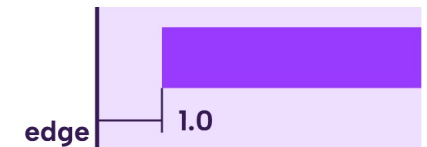
Only these. Any other hole size fails DRC.



Via

0.7 / 1.9 mm

Hole / pad. Set as the net class default.



Copper to edge

1.0 mm

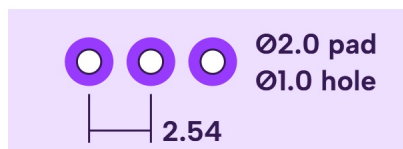
From the Edge.Cuts outline to any copper.



Slot width

1.0 mm

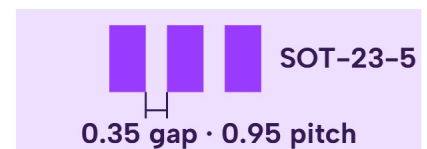
1.0 mm router, so inside corners are R 0.5.



2.54 mm headers

1.0 / 2.0 mm

Hole / pad, 0.54 mm gap. Enlarge stock pads.



Finest SMD pitch

0.95 mm

SOT-23-5 and SOIC pass. No TSSOP or QFN.